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Prototype Implementation of a Photoplethysmography System for Early Arrhythmia Detection

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I. Introduction / Brief Description

This prototype focuses on the early detection of arrhythmia using a Photoplethysmography (PPG)-based monitoring system. Arrhythmia, or irregular heartbeat, is a condition that can lead to serious cardiovascular complications if not identified at an early stage. Early detection is therefore essential to reduce risks and enable timely medical intervention.

The system utilizes a MAX30102 optical sensor paired with an Arduino Nano microcontroller to capture PPG signals. These signals reflect blood volume changes in the microvascular bed of tissue, which can be analyzed to identify irregular heart rhythms.

At this stage, the team has successfully gathered initial PPG data and confirmed that the sensor is functional. However, the collected signals still contain noticeable noise, which highlights the need for further filtering and signal processing to improve accuracy and reliability.

Beyond the technical aspect, this prototype is designed to be affordable and accessible, making it practical for use in the Philippines, especially in rural areas where access to advanced healthcare facilities is limited. By providing a low-cost solution, the system can also benefit individuals who require immediate health response or continuous monitoring for early detection of heart irregularities. The affordability of the device ensures that it can reach a wider population, supporting preventive healthcare and potentially reducing the burden of cardiovascular diseases.

II. Objectives

To design and implement a prototype capable of the early detection of arrhythmia using PPG signals, providing a low-cost and accessible monitoring tool that can benefit communities, particularly in rural areas of the Philippines where access to advanced healthcare technologies is limited.

- To integrate the MAX30102 sensor with the Arduino Nano microcontroller for real-time PPG signal acquisition.
- To develop and code the program using the Arduino IDE for efficient data collection and processing.
- To collect and analyze PPG data for identifying irregular heartbeat patterns associated with arrhythmia.
- To address noise issues in the raw data through filtering and signal processing techniques, improving accuracy and reliability.
- To evaluate the prototype's potential for affordable early arrhythmia detection, ensuring it can be realistically deployed in resource-limited settings.

Cardiovascular diseases remain the leading cause of death in the Philippines, and early detection tools are critical for prevention and timely treatment. Affordable, portable devices can help bridge the gap in healthcare access, especially in rural areas where diagnostic equipment is scarce (World Health Organization, 2021; Department of Health Philippines, 2022).

III. Materials and Components

Arduino Nano Microcontroller

Low-power Wi-Fi and Bluetooth-enabled microcontroller

Operates at 3.3V

MAX30102 Pulse Oximeter & Heart-Rate Sensor

Integrated red and IR LEDs with photodetector

Used for PPG signal acquisition

Wires / Jumper Cables

For sensor-to-microcontroller connections

Breadboard

To ensure hardware connection.

IV. Methodology / Step-by-Step Procedure

Stage 1: Component Preparation

The components required for the development of the photoplethysmography (PPG) device include the Arduino Nano microcontroller, MAX30102 pulse oximeter sensor, connecting wires, breadboard, and a custom-designed 3D-printed case. Each component was tested to ensure functionality prior to integration.

Stage 2: Hardware Connection

The Arduino Nano microcontroller was interfaced with the MAX30102 sensor module using the I²C communication protocol. Specifically, the SCL (Serial Clock Line) and SDA (Serial Data Line) pins of the sensor were connected to their corresponding terminals on the Arduino Nano. The GND pin of the sensor was linked to the microcontroller's ground, while the VIN pin was connected to the 3.3V power supply output. The system was served by a direct voltage source from the pc, providing stable voltage output. All connections were established using jumper wires on a breadboard to ensure proper alignment, secure contact, and minimized electrical noise.

Stage 3: 3D Case Integration

A custom 3D-printed case was designed to house the assembled components. The enclosure was modeled to fit the Arduino Nano, MAX30102 sensor, and battery with precision while allowing access to external ports for programming. The 3D case also provided mechanical protection and enhanced device portability, ensuring stable sensor placement during operation.

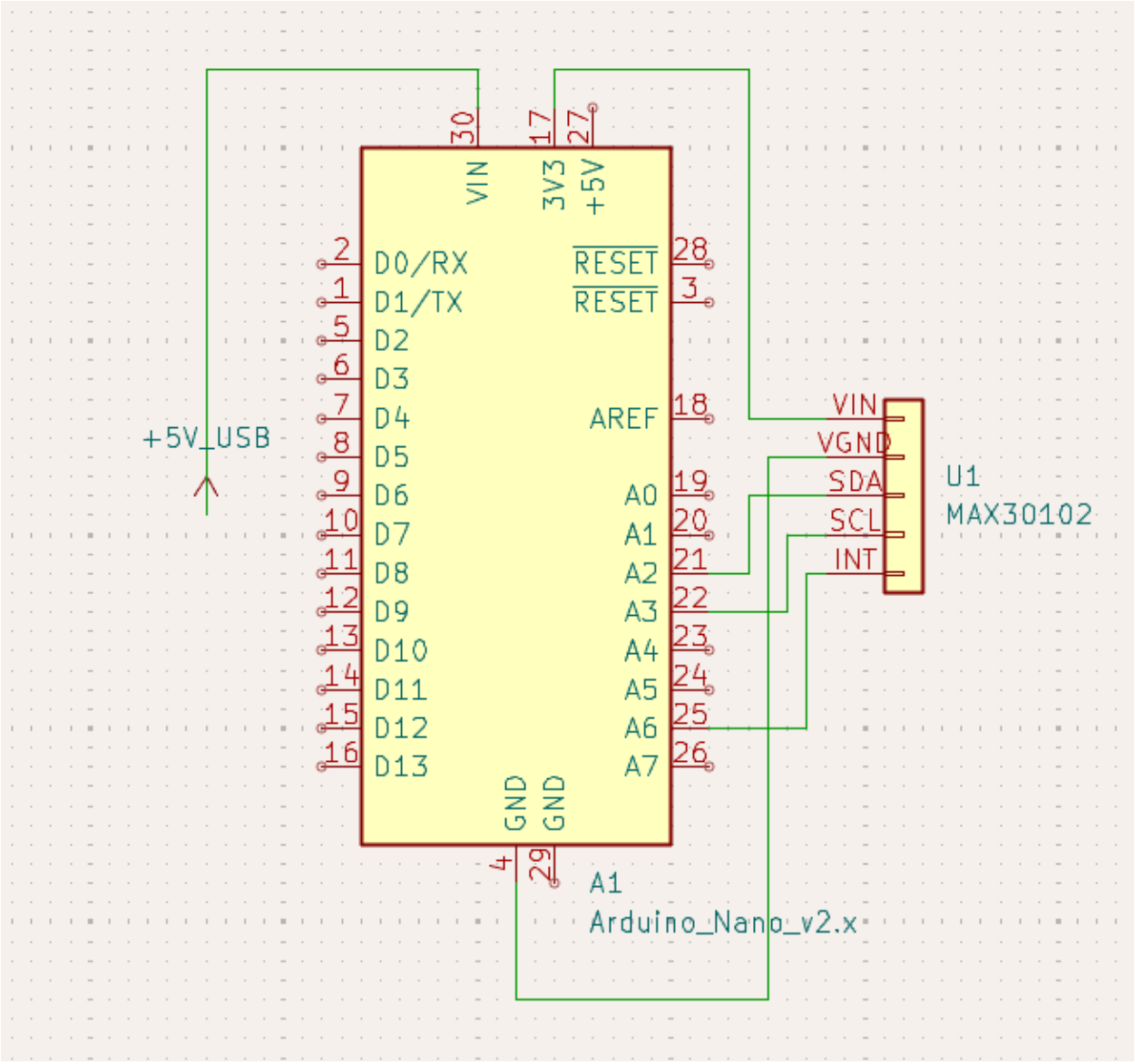
Stage 4: Software Configuration and Data Acquisition

Programming of the Arduino Nano was performed using the Arduino Integrated Development Environment (IDE). The uploaded program enabled the microcontroller to read optical signals from the MAX30102 sensor, process the data, and transmit it for display and analysis. Real-time data visualization was achieved through serial communication with a computer interface, allowing verification of heart rate and oxygen saturation readings.

Stage 5: System Evaluation and Testing

The fully assembled PPG device was subjected to a series of functional tests to assess accuracy, stability, and signal reliability. The obtained PPG waveforms were analyzed and compared with standard reference signals to validate the device's performance. Observations from the evaluation stage were used to confirm the system's capability to detect and represent pulse signals effectively.

V. Circuit Diagrams / Schematics



VI. Program / Code

Arduino IDE

```
#include <Wire.h>
#include "MAX30105.h"
#include "heartRate.h"

MAX30105 particleSensor;

const byte RATE_SIZE = 4;
byte rates[RATE_SIZE];
byte rateSpot = 0;
long lastBeat = 0;

float beatsPerMinute;
int beatAvg;

void setup()
{
  Serial.begin(115200);
  delay(1000);

  Serial.println("Initializing sensor...");

  if (!particleSensor.begin(Wire, I2C_SPEED_FAST))
  {
    Serial.println("MAX30105 was not found. Please check wiring/power.");
    while (1);
  }

  byte ledBrightness = 0x1F;
  byte sampleAverage = 4;
  byte ledMode = 2;
  int sampleRate = 100;
  int pulseWidth = 411;
  int adcRange = 4096;

  particleSensor.setup(ledBrightness, sampleAverage, ledMode, sampleRate, pulseWidth,
adcRange);
```

```
  Serial.println("Place your finger on the sensor.");
  delay(2000);
}

void loop()
{
  long irValue = particleSensor.getIR();

  if (checkForBeat(irValue) == true)
  {
    long delta = millis() - lastBeat;
    lastBeat = millis();

    beatsPerMinute = 60 / (delta / 1000.0);

    if (beatsPerMinute < 255 && beatsPerMinute > 20)
    {
      rates[rateSpot++] = (byte)beatsPerMinute;
      rateSpot %= RATE_SIZE;

      beatAvg = 0;
      for (byte x = 0 ; x < RATE_SIZE ; x++)
        beatAvg += rates[x];
      beatAvg /= RATE_SIZE;
    }
  }
  Serial.print(irValue);
  Serial.print(",");
  Serial.print(beatsPerMinute);
  Serial.print(",");
  Serial.print(beatAvg);
  Serial.print(",");
  Serial.print(irValue < 50000 ? 0 : 1);
  Serial.println();
}
```

VII. Testing and Results

Initial testing confirmed that the PPG sensor works and produces readable signals.

However, the data is noisy, which may affect detection accuracy.

Further testing will focus on applying filters and algorithms to clean the signal.

	A	B	C	D	E	F
1	#!	Workbook	Custom	Custom		
2	#!	To play this data back: Open the original Excel workbo				
3	#!	For more information visit http://aka.ms/hackingstem				
4	118836	50.13	43	1		
5	118857	50.13	43	1		
6	118855	50.13	43	1		
7	118889	50.13	43	1		
8	118889	50.13	43	1		
9	118718	50.13	43	1		
10	118488	50.13	43	1		
11	118459	50.13	43	1		
12	118534	50.13	43	1		
13	118587	50.13	43	1		
14	118631	50.13	43	1		
15	118606	50.13	43	1		
16	118564	50.13	43	1		
17	118544	50.13	43	1		

VIII. Analysis and Discussion

The prototype has been successfully tested for its capabilities and limits and to check if both the core parts- the MAX30102 sensor and the Arduino Nano microcontroller works.While the raw data confirmed its functionality, noise was observed which affected the clarity of the data collection of the calculated heart rate values. This highlights the importance of applying digital filtering techniques, such as moving average or bandpass filters, to improve signal quality.

Despite these limitations, the system showed potential as a low-cost and portable tool for early arrhythmia detection. Its affordability and accessibility make it particularly relevant in rural healthcare settings in the Philippines, where advanced diagnostic equipment is often unavailable. Further refinement in signal processing and algorithm development will be essential to enhance accuracy and reliability for real-world application.

Another important observation was the variability in readings depending on sensor placement and user conditions. Consistent finger positioning and stable contact with the sensor were critical to obtaining reliable signals. The integration of a custom 3D-printed case will help reduce movement and improve stability, but further ergonomic design improvements could enhance usability in real-world applications and the design and testing is still ongoing.

IX. Conclusion and Recommendations

The prototype successfully demonstrated the integration of the MAX30102 sensor with the Arduino Nano microcontroller to acquire and analyze PPG signals. Initial testing confirmed that the system can capture heart rate data, but noise and signal instability remain significant challenges. Despite these limitations, the project shows strong potential as a low-cost, portable, and accessible solution for early arrhythmia detection, particularly in rural communities where advanced healthcare facilities are scarce.

Recommendations and further enhancements are open to improve this prototype, as such:

- Implement digital filtering techniques such as moving average (accelerometer), bandpass, or adaptive filters to reduce noise and improve signal quality.
- Improve the mechanical design of the 3D-printed case to ensure stable sensor placement and minimize motion factors.